

Homework 4

For this homework you will work with your group on a **GitHub** repo that creates a nice website to answer the questions below. One group member needs to turn in the link to this website!

Assignment Tasks

A `.qmd` file that creates a final website is the final product. I would recommend each group member create their own `.qmd` file for their version of the solutions. Then have the group decide on how to merge the solutions for the final file. You don't have to follow this structure though!

- In the final document, answers to the questions below along with a discussion of what you are doing (via markdown) and comments in the code should be included.
- Use **tidyverse functions and manipulations for most aspects of this homework.**

Task 1 - Querying an API and a Tidy-Style Function

For this section, you'll connect to the news API here: newsapi.org. You'll need to go to register for a key at that web site!

1. Use `GET()` from the `httr` package to return information about a topic that you are interested in that has been in the news lately (store the result as an R object). Note: We can only look 30 days into the past with a free account.

```
library(httr)
GET("https://newsapi.org/v2/everything?q=Iran&from=2026-05-02&language=en&
    pageSize=50&apiKey=myKeyGoesHere")
```

2. Parse what is returned and find your way to the data frame that has the actual article information in it (check `content`). Note the first column should be a list column!
3. Now write a quick function that allows the user to easily query this API. The inputs to the function should be the title/subject to search for (string), a time period to search from (string - you'll search from that time until the present), and an API key.

Use your function twice to show it works!

Task 2 - Summarizing Roller Coaster Data

For this part, we'll use data about roller coasters. The data is about different aspects of roller coasters and is available at <https://www4.stat.ncsu.edu/online/datasets/635USrollercoasters.csv>

The variables are

- Coaster - Name of roller coaster
- Park - Amusement park where roller coaster is located
- City
- State
- Census Region
- Year Opened
- AgeGroup - 1 (Older), 2 (Recent), or 3 (Newest)
- TopSpeed - in mph
- MaxHeight - in feet
- Scale - Extreme, Thrill, Family
- Type - Steel, Wood
- Design - Sit Down, Inverted, Bobsled, etc.
- Drop - in feet
- Length - in feet
- Duration - in seconds
- Inversions
- Number of Inversions
- Make
- Latitude
- Longitude
- Boundary - for plotting on maps

No need to use the `Boundary` variable or the `Latitude` and `Longitude` variables in your EDA below!

- Read in this data and convert variables to factors as needed.

Basic EDA

We've talked about the general process of conducting an EDA. You try to understand how your data is stored, what is missing, and you try to summarize the variables both numerically and visually to understand relationships within the data.

- Look at how the data is stored and see if everything makes sense.
- Document the missing values in the data.

Categorical variables

- Create a one-way contingency table, a two-way contingency table, and a three-way contingency table for some of the factor variables you created previously. Use `table()` to accomplish this.
 - Interpret a number from each resulting table (that is, pick out a value produced and explain what that value means.)
- Create a conditional two-way table using `table()`. That is, condition on one variable's setting and create a two-way table. Do this using two different methods:
 - Once, by subsetting the data (say with `filter()`) and then creating the two-way table
 - Once, by creating a three-way table and subsetting it
- Create a two-way contingency table using `group_by()` and `summarize()` from `dplyr`. Then use `pivot_wider()` to make the result look more like the output from `table()`.
- Create a stacked bar graph and a side-by-side bar graph. Give relevant `x` and `y` labels, and a title for the plots.

Numeric variables (and across groups) Consider the numeric variables:

- Find measures of center and spread for two of these variables
 - Repeat while subsetting the data in a meaningful way.
- Find measures of center and spread across a single grouping variable for two of these variables
- Find measures of center and spread across two grouping variables for two of these variables
- Create a correlation matrix between all of the numeric variables
- Create a histogram, kernel density plot, and boxplot for two of the numeric variables across one of the categorical variables (that is, create graphs that can compare the distributions across the groups on the same plot (no *faceting* here)). Add appropriate labels and titles.
- Create two scatterplots relating numeric variables. Color the points by a categorical variable in each. Add appropriate labels and titles.
- Repeat the scatter plot step but use faceting to obtain graphs at each setting of another categorical variable.
- Repeat the scatter plot step but use faceting to obtain graphs at each combination of two categorical variables.

After each summary or graph, you should discuss what is interesting about it or what it tells you!

To Turn In

One member should submit the link to the nicely rendered site on Moodle!